



**AIMS**

African Institute for  
Mathematical Sciences  
**RWANDA**

# Introduction to Probability and Statistics

## Lecture 10: Convergence of the mean, Central Limit Theorem

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1 The statistic  $\overline{X}_n$

2 Central Limit Theorem

## WLLN

## Weak Law of Large Number (WLLN)

Let  $X_1, \dots, X_n$  be  $n$  independent and identically distributed (iid) r.v.'s such that  $E(X_i) = a < +\infty$  and  $Var(X_i) = b$ ,  $i = 1, \dots, n$ .

Then  $\bar{X}_n$  **converge in probability** to  $a$ .

**Proof:** Apply Chebyshev's inequality

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**Proof:** Apply Chebyshev's inequality

$$\forall \varepsilon > 0, P(|\bar{X}_n - a| > \varepsilon) \leq \frac{b}{n \varepsilon^2} \xrightarrow{n \rightarrow +\infty} 0$$

$$\iff \forall \varepsilon > 0, \lim_{n \rightarrow +\infty} P(|\bar{X}_n - a| > \varepsilon) = 0$$

$$\iff \bar{X}_n \xrightarrow{P} a.$$

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- $\implies$  Convergence in Probability:  $\implies \bar{X}_n \xrightarrow{P} a$ .

## Limit of $\overline{X}_n$ m.g.f

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Limit of  $\bar{X}_n$  m.g.f**Proof of the first point:**

$$M_{\bar{X}_n}(t) = E(e^{t\bar{X}_n}) = E\left(e^{\frac{t}{n} \sum_{i=1}^n X_i}\right) = \prod_{i=1}^n E(e^{\frac{t}{n} X_i}) = \left[E(e^{\frac{t}{n} X_i})\right]^n$$

But  $e^{\frac{t}{n} X_i} = 1 + \frac{t}{n} X_i + \frac{t^2}{n^2} X_i^2 + \dots$

Then  $E(e^{\frac{t}{n} X_i}) \sim \left[1 + \frac{t}{n} E(X_i) + o\left(\frac{t}{n}\right)\right]^n = \left[1 + \frac{ta}{n} + o\left(\frac{t}{n}\right)\right]^n$

where  $o$  is a function such that  $\lim_{h \rightarrow 0} o(h)/h = 0$ .

$$\begin{aligned} \left[1 + \frac{ta}{n} + o\left(\frac{t}{n}\right)\right]^n &= \exp\left\{n \log\left[1 + \frac{at + n o(t/n)}{n}\right]\right\} \\ &= \exp\left\{n \times \frac{at + n o(t/n)}{n}\right\} \\ &= \exp\{at + n o(t/n)\} \end{aligned}$$

which goes to  $\exp\{at\}$  when  $n$  goes to infinity since  $n o(t/n)$  goes to 0.

Hence

$$M_{\bar{X}_n}(t) \xrightarrow{n \rightarrow +\infty} e^{ta}.$$

## Convergence in distribution

$$\implies \bar{\mathbf{X}}_n \xrightarrow{d} \mathbf{X}, \text{ where } X \sim \delta_a.$$

## Convergence in distribution

$Z_n$  converges **in distribution** or **in law** if,

$$\lim_{n \rightarrow +\infty} F_{Z_n}(z) = F_Z(z),$$

at each point  $z$  where  $F_Z$  is continuous.

Notation:  $Z_n \xrightarrow{d} Z$  or  $Z_n \xrightarrow{\mathcal{L}} Z$

## Point mass distribution

$$\Rightarrow \bar{X}_n \xrightarrow{d} X, \text{ where } \mathbf{X} \sim \delta_{\mathbf{a}}.$$

## Point mass distribution

A r.v.  $X$  on  $\mathbb{R}$ , is said to follow the **point mass distribution**  $\delta_c$ , if  $P(X = c) = 1$ .

The c.d.f of  $X$  is defined as:

$$F(x) = \begin{cases} 0 & x < c \\ 1 & x \geq c \end{cases}$$

Moment generating function:

$$M_X(t) = E(e^{tX}) = e^{tc} P(X = c) = e^{tc}.$$

Convergence in distribution  $\overline{X}_n$ 

## Continuity Theorem for m.g.f.'s

Let  $Z_1, Z_2, \dots$  be a sequence of r.v.'s,  $F_n$  be the c.d.f of  $Z_n$  and  $M_{Z_n}(t)$  the m.g.f of  $Z_n$ . Then:

$$\left( M_{Z_n}(t) \xrightarrow{n \rightarrow +\infty} M_Z(t) \right) \iff \left( Z_n \xrightarrow{d} Z \right).$$

## Proposition

Let  $X_1, X_2, \dots$  be a sequence of r.v.'s, with  $F_n$ , the c.d.f of  $X_n$ . Let  $X$  be a r.v with the point mass distribution  $\delta_c$ . Then:

$$\left( X_n \xrightarrow{d} X \right) \iff \left( X_n \xrightarrow{P} c \right).$$

## Proof of the proposition

- Suppose  $\lim_{n \rightarrow +\infty} F_{X_n}(x) = F_X(x)$  with  $X \sim \delta_c$ .

$$\begin{aligned}\forall \varepsilon > 0, P(|X_n - c| > \varepsilon) &= P(X_n > c + \varepsilon) + P(X_n < c - \varepsilon) \\ &= 1 - F_n(c + \varepsilon) + F_n(c - \varepsilon)\end{aligned}$$

$$\lim_{n \rightarrow +\infty} F_{X_n}(c - \varepsilon) = F_X(c - \varepsilon) = 0 \text{ and } \lim_{n \rightarrow +\infty} F_{X_n}(c + \varepsilon) = F_X(c + \varepsilon) = 1.$$

Hence  $X_n \xrightarrow{P} c$ .

- Suppose that  $\forall \varepsilon > 0, P(|X_n - c| > \varepsilon) \rightarrow 0$  when  $n \rightarrow +\infty$ .

Then  $\lim_{n \rightarrow +\infty} [P(X_n > c + \varepsilon) + P(X_n < c - \varepsilon)] \rightarrow 0$  which means that

$$\lim_{n \rightarrow +\infty} P(X_n > c + \varepsilon) = 0 \text{ and } \lim_{n \rightarrow +\infty} P(X_n < c - \varepsilon) = 0.$$

Therefore

$$\forall \varepsilon > 0, \lim_{n \rightarrow +\infty} F_{X_n}(c - \varepsilon) = 0 \text{ and } \lim_{n \rightarrow +\infty} F_{X_n}(c + \varepsilon) = 1.$$

Hence  $\lim F_n(x) = F_X(x)$  such that  $F_X(x) = 1$  for  $x \geq c$  and  $F_X(x) = 0$  for  $x < c$  which means that  $X_n$  converges to  $X$  having a point mass probability  $\delta_c$ .

## CLT

## Central Limit Theorem (CLT)

Let  $X_1, \dots, X_n$  be  $n$  independent and identically distributed (iid) r.v.'s with  $E(X_1) = \mu$  and  $\text{Var}(X_1) = \sigma^2 > 0$  (both finite).

Then:

$$\sqrt{n} \left( \frac{\bar{X}_n - \mu}{\sigma} \right) \sim \mathcal{N}(0, 1) \text{ when } n \rightarrow +\infty.$$

The r.v.  $Z_n = \sqrt{n}(\bar{X}_n - \mu)/\sigma$  converges **in distribution** (or in law) to a standard normal r.v. or  $X_n$  behaves as a normal distribution with parameters  $(\mu, \sigma^2/n)$  when  $n$  goes to infinity.

$$\lim_{n \rightarrow +\infty} F_{Z_n}(z) = \Phi(z)$$



## CLT

**Proof:** We are going to show that  $M_{Z_n}(t) \xrightarrow{n \rightarrow +\infty} M_Z(t)$ .

$$Z_n = \sqrt{n} \left( \frac{\bar{X}_n - \mu}{\sigma} \right) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left( \frac{X_i - \mu}{\sigma} \right)$$

Let  $Y_i = \frac{X_i - \mu}{\sigma}$ ,  $Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n Y_i$ .

$$M_{Z_n}(t) = E(e^{tZ_n}) = \prod_{i=1}^n E(e^{\frac{t}{\sqrt{n}} Y_i}) = \left[ E(e^{\frac{t}{\sqrt{n}} Y_i}) \right]^n.$$

$$e^{\frac{t}{\sqrt{n}} Y_i} \approx 1 + \frac{t}{\sqrt{n}} Y_i + \frac{t^2}{2n} Y_i^2 \iff E(e^{\frac{t}{\sqrt{n}} Y_i}) \approx 1 + \frac{t}{\sqrt{n}} E(Y_i) + \frac{t^2}{2n} E(Y_i^2)$$

Then  $E(e^{\frac{t}{\sqrt{n}} Y_i}) \approx 1 + \frac{t^2}{2n}$  since  $E(Y_i) = 0$  and  $\text{Var}(Y_i) = E(Y_i^2) = 1$ .

$$\left[ E(e^{\frac{t}{\sqrt{n}} Y_i}) \right]^n \approx \left( 1 + \frac{t^2}{2n} \right)^n.$$

But  $\left( 1 + \frac{t^2}{2n} \right)^n = e^{n \log(1 + \frac{t^2}{2n})} \approx e^{\frac{t^2}{2}}$  since  $\log(1 + \frac{t^2}{2n}) \approx \frac{t^2}{2n}$  when  $n$  large.

Therefore  $\lim_{n \rightarrow +\infty} M_{Z_n}(t) = e^{\frac{t^2}{2}}$  which is the m.g.f. of a standard normal distribution.

**Conclusion:**  $Z_n \xrightarrow{d} Z$ , where  $Z \sim \mathcal{N}(0, 1)$ .

## CLT

**Example:**

Let  $X_1, \dots, X_n$  be independent Bernoulli r.v.'s with parameter  $p$ .

We have  $E(X_i) = p$  and  $\text{Var}(X_i) = p(1 - p)$ .

Applying the CLT leads to :

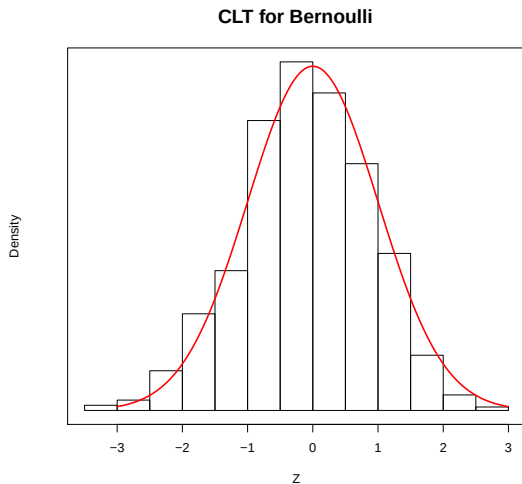
$$\sqrt{n} \left( \frac{\bar{X}_n - p}{\sqrt{p(1 - p)}} \right) \sim \mathcal{N}(0, 1).$$

**Simulation:**

- Choose a value for  $p$  e.g.  $p = 0.8$ .
- Draw 100 realizations of a Bernoulli with parameter 0.8.
- Compute  $\bar{X}_n$ , the mean of the 100 values.
- Reproduce this experiment 1000 times to obtain a sample of  $\bar{X}_n$ .
- Build the histogram to investigate the distribution of the variables

$$\sqrt{n}((\bar{X}_n - p)/\sqrt{p(1 - p)}).$$

# Illustration of the CLT



## R code

```
p<-0.8 # Choose a value for p
n<-100 # Size of the sample
# Generate a sample of n Bernoulli r.v.'s realizations
x<-sample(c(0,1), n, replace=TRUE, prob=c(0.2,0.8))
M<-1000 # Number of samples
#  $Z = \sqrt{n}(\bar{X}_n - p) / \sqrt{p(1-p)}$ 
Z<-rep(0,M) # Initialization
for (i in 1:M)
Z[i]<-sqrt(n)*((mean(rbinom(n,1,p))-p)/sqrt(p*(1-p)))
# Histogram
hist(Z, nclass=20, probability=T, main=list("CLT for
Bernoulli",cex=1.5),
ylab="Density", yaxt = "n")
# Adding the p.d.f.
z<-seq(-3,3,0.01)
gaussz<-dnorm(z,0,1)
lines(z,gaussz)
box() # To add a frame
```